AWS Assignment Solutions

B.Tech CSE Level — Cloud Computing

Q1. Cloud Services Used Weekly — Classification

Find three cloud services that you personally use every week. Classify each one as IaaS, PaaS, or SaaS. If a service could fit into more than one category depending on the use, explain why.

Cloud Service	Service Model	Reason
Google Drive	SaaS	Google provides a complete cloud storage application. Users simply upload, download, and share files without managing servers or software.
GitHub	SaaS (Primary), PaaS (for GitHub Actions)	GitHub is mainly a SaaS platform for source code hosting and collaboration. However, GitHub Actions provides a platform to automatically build, test, and deploy applications, making it similar to PaaS.
Firebase	PaaS	Firebase provides backend services such as authentication, databases, cloud storage, and hosting. Developers focus on application development without managing infrastructure.

Conclusion:

Cloud services can sometimes belong to more than one service model depending on how they are used. For example, GitHub is primarily SaaS but also offers PaaS capabilities through GitHub Actions.

Q2. Online Coding Contest for 25,000 Participants (Advitya)

VIT Bhopal University wants to launch an online coding contest for 25,000 participants during Advitya. Decide:

(a) Deployment Model — Public Cloud

- Supports a very large number of users.
- Easily scales resources during peak traffic.
- Cost-effective since resources are rented only when needed.
- Major cloud providers like AWS, Azure, and Google Cloud provide reliable infrastructure.

(b) Service Model — Platform as a Service (PaaS)

- Developers can quickly deploy the coding platform.
- Infrastructure management is handled by the cloud provider.
- Allows focus on application development instead of server maintenance.

(c) Most Critical Cloud Characteristic — Rapid Elasticity

- Thousands of students may log in simultaneously.
- Resources should automatically increase during the contest and decrease afterward.

- Ensures smooth performance without server crashes.

Q3. Startup Deployment Model Recommendation

A startup has ■3 lakh budget, two developers, must launch in one month, and customer data is confidential. Recommend ONE deployment model and justify why the other three are less suitable.

Recommended Deployment Model: Hybrid Cloud

- Sensitive customer data can be stored securely in a private cloud.
- Public cloud resources can host the web application and handle user traffic.
- Offers a balance between security, scalability, and cost.
- Suitable for startups with limited resources.

Why the Other Deployment Models Are Less Suitable

Private Cloud

- High infrastructure and maintenance cost.
- Requires dedicated IT staff.
- Not suitable for a small startup with only two developers.

Public Cloud

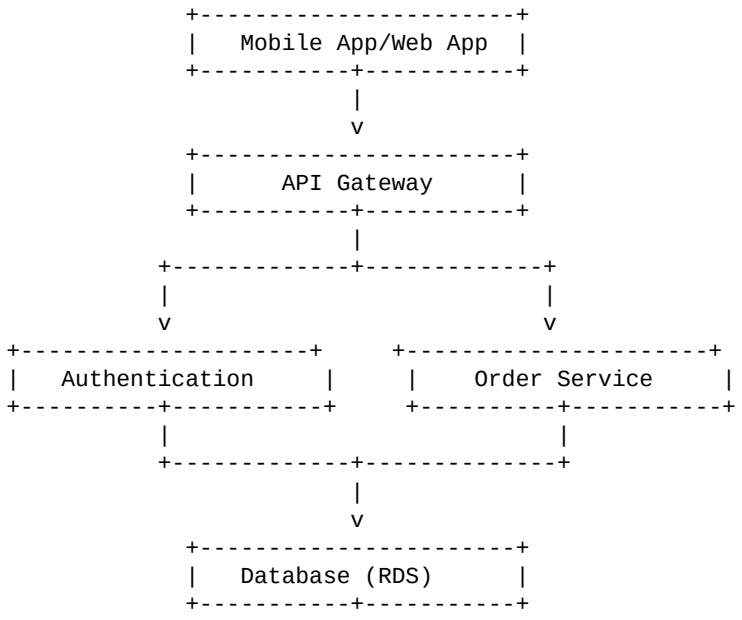
- Very cost-effective and scalable.
- However, storing confidential customer data entirely on a public cloud may raise security and compliance concerns.

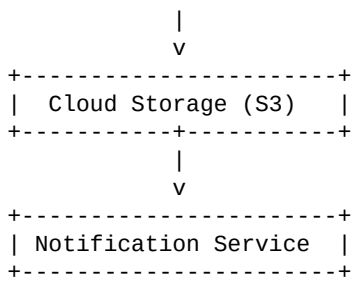
Community Cloud

- Designed for organizations with similar objectives (such as universities or government agencies).
- Not appropriate for a private startup business.

Q4. Cloud Architecture — Food Delivery App

Draw your own cloud architecture for a food delivery app (use only boxes and arrows). Explain why you placed each component where you did.





Explanation

Mobile App/Web App

Used by customers to browse restaurants and place orders.

API Gateway

- Acts as the single entry point for all user requests.
- Improves security and request routing.

Authentication Service

Verifies user identity before allowing access.

Order Service

Processes food orders and communicates with the database.

Database (Amazon RDS)

Stores customer information, restaurant details, and order history.

Cloud Storage (Amazon S3)

Stores food images, restaurant logos, invoices, and receipts.

Notification Service

Sends SMS, email, or push notifications for order updates.

Q5. Ranking Cloud Characteristics — Online Examination Platform

Rank the following cloud characteristics from most important to least important for an online examination platform.

Rank	Cloud Characteristic	Reason
1	Rapid Elasticity	Handles thousands of students logging in simultaneously during exams.
2	Broad Network Access	Students can access the platform from laptops, tablets, or smartphones anywhere with internet access.
3	On-demand Self Service	Administrators can quickly create exams, allocate resources, and manage users without manual intervention.
4	Resource Pooling	Efficiently shares computing resources among many users, reducing costs.
5	Measured Service	Important for monitoring resource usage and billing, but it has little direct impact on students during an examination.

Why Rapid Elasticity Is the Most Important

During an online examination, thousands of students may access the platform at the same time. Rapid Elasticity automatically allocates additional computing resources during peak demand and releases them afterward. This prevents server overload, reduces downtime, and ensures a smooth examination experience.

Why Measured Service Is the Least Important

Measured Service tracks and reports resource consumption for billing and monitoring purposes. Although it helps administrators optimize costs, it does not directly improve the performance, availability, or reliability of the examination platform. Therefore, it is the least critical characteristic in this scenario.